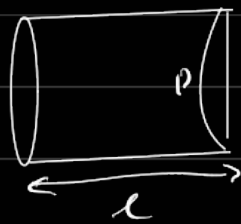


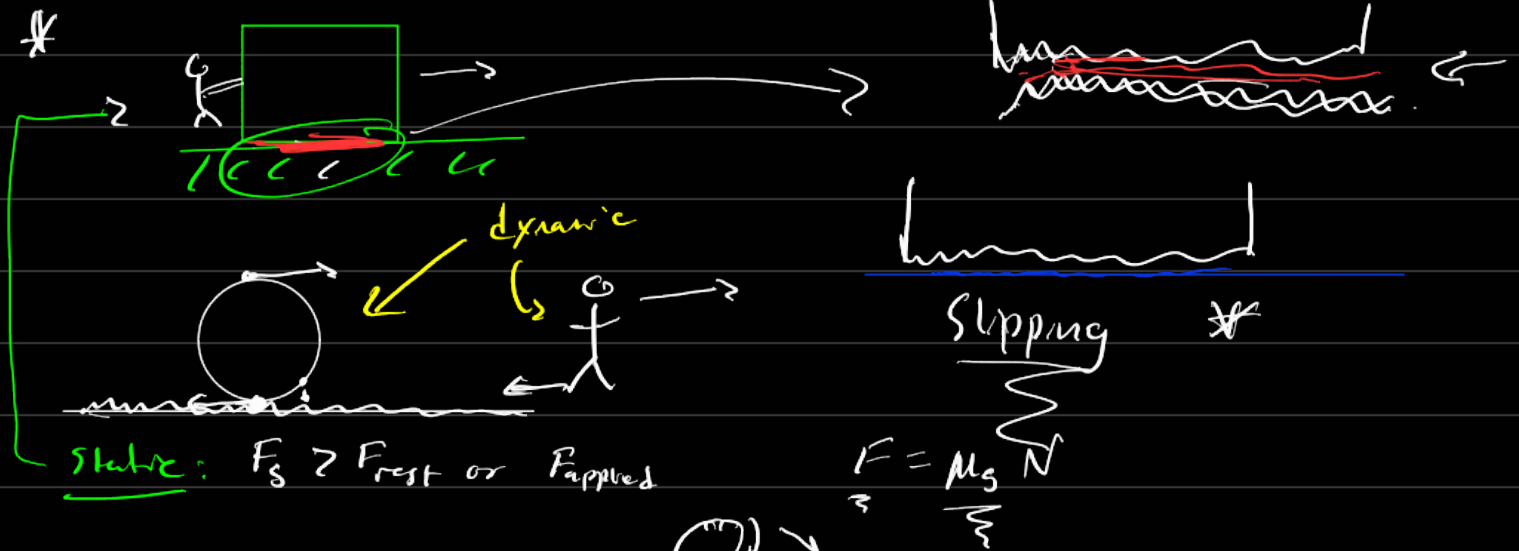
②



$$[X] = \frac{\text{Vol} \sim m^3}{\Delta t}, \quad X = X(r, \frac{p}{e}, n)$$

$$[n] = N s m^{-2}$$

$$[\frac{p}{e}] =$$



③ Rolls: dynamic:  
from A to C  
from C to D: no rolling

Rolling: get moment of inertia (K.E.): add a rotational component

A to B:  $mgh = \underbrace{\frac{1}{2} m v_B^2}_{\text{motion}} + \underbrace{\frac{1}{2} I \omega^2}_{\text{rotational}}$

$$I_{\text{ring}} = R^2 M, \quad \omega = v/R$$

$$(K.E.)_{\text{rot}} = \frac{1}{2} m v^2$$

$$mgh = m v_B^2 \Rightarrow \boxed{v_B = \sqrt{gh}}$$

\* Suppose:  $(A \rightarrow C) \rightarrow (C \rightarrow A)$ : replace  $h \rightarrow 0.1h$

$$\tilde{v}_B = \sqrt{g \cdot 0.1h}$$

$$\Rightarrow \Delta(K.E.) = \frac{1}{2} mgh - \frac{1}{2} mg \cdot 0.1h$$

$$= \frac{1}{2} mg \cdot 0.1h$$

\* Work:  $\Delta K.E. = \frac{1}{2} mg \cdot 0.1h$

→ Here, this is the func. work → en. loss

$$* \text{ at C : } \underbrace{(mgh(0.85))}_{\text{Pot.}} - \underbrace{\frac{1}{2}mgh(0.25)}_{\text{En. loss}}$$

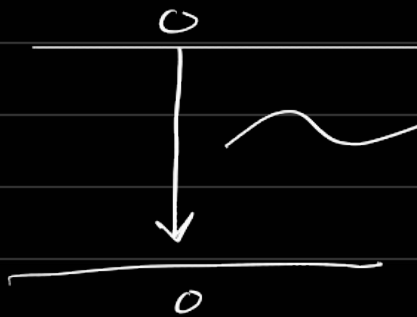
$$\rightarrow mgh(0.85), \quad \underline{V_c = 0}$$

$$E_c = mgh(0.85)$$

$$E_D = E_c \Rightarrow mgh(0.85) = \frac{1}{2} m V_D^2$$

$$\boxed{V_D^2 = 2(0.85)gh}$$

(4)



$$\gamma: \lambda = 502.0 \text{ nm}$$

$$\Delta\lambda = 2.1 \times 10^{-3} \text{ nm}$$

find  $\Delta t$

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

$$* N(t) = N_0 e^{-t/\tau} \quad (\text{Decay-rate})$$

$$\boxed{E = \frac{hc}{\lambda}}$$

$$f \cdot \lambda = c$$

$$f = \frac{c}{\lambda}$$

taking der.

$$\frac{dE}{d\lambda} = -\frac{hc}{\lambda^2} \rightarrow \frac{\Delta E}{\Delta\lambda} = -\frac{hc}{\lambda^2} \Rightarrow$$

$$\Delta E = \left| \frac{hc}{\lambda^2} \right| \Delta\lambda$$

$$\Delta E \Delta t \approx \frac{\hbar}{2} \rightarrow \frac{hc}{\lambda^2} \Delta\lambda \Delta t \approx \frac{\hbar}{2}$$

$$\hbar = \frac{h}{2\pi}$$

$$\boxed{\Delta t \approx \frac{\hbar \lambda^2}{2hc \Delta\lambda} = \frac{\lambda^2}{2(2\pi)c \Delta\lambda}}$$

$$\lambda = 502.0 \text{ nm}$$

$$\Delta\lambda = 2.1 \times 10^{-3} \text{ nm}$$

